

Function Notation Review

Let $f(x) = 5x^2$, $g(x) = \frac{x-3}{x+1}$, and $h(x) = \sqrt{x+2}$
 Evaluate:
 1. $f(-2)$ 2. $g(4)$ 3. $h(7)$

What is function notation?

Using $f(x)$ instead of y : $y = 3x + 4 \Rightarrow f(x) = 3x + 4$
 "f of x" $\neq$ Does not mean multiply

- Can include other variables and function names:

Function Rule

$$f(x) = 3x + 4$$

$$g(x) = x^2 - 8$$

$$h(a) = \sqrt{a+3}$$

Evaluating

$$f(2) = 3(2) + 4 = 10$$

$$g(5) = (5)^2 - 8 = 17$$

$$h(6) = \sqrt{6+3} = \sqrt{9} = 3$$

$$g(-2) = -4$$

$$= (-2)^2 - 8 = -4$$

Always use () when subbing in.

Operations with Functions: How to add, subtract, multiply and divide with functions

The Rules:

Adding $(f + g)(x) = f(x) + g(x)$

★ Subtracting $(f - g)(x) = f(x) - g(x)$ Be careful @ distributive

Multiplying $(f \cdot g)(x) = f(x) \cdot g(x)$

Dividing $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$; $g(x) \neq 0$

Dividing by zero is undefined

• means multiply
 • does not mean mult.

Ex2 $f(x) = 2x^2$ $g(x) = 2x - 2$ Find the following:

1. $(f + g)(3) =$

$$f(3) + g(3)$$

$$2(3)^2 + 2(3) - 2$$

$$18 + 6 - 2$$

$$\boxed{22}$$

2. $(f - g)(-4) =$

$$f(-4) - (g(-4))$$

$$2(-4)^2 - (2(-4) - 2)$$

$$32 - (-10)$$

$$\boxed{42}$$

3. $(fg)(-1) =$

$$f(-1) \cdot g(-1)$$

$$[2(-1)^2][2(-1) - 2]$$

$$(2)(-4)$$

$$\boxed{-8}$$

4. $\left(\frac{f}{g}\right)(x) =$

$$\frac{f(x)}{g(x)} = \frac{2x^2}{2x - 2}$$

$$= \frac{x^2}{x - 1}$$

$x - 1 \neq 0$
 $x \neq 1$

more than one

What is a composite function (a.k.a. $\Rightarrow$ function composition)?

When we put one function inside of another. function

- Notation: Easiest! output of $f(g(x))$ but also written as $(f \circ g)(x)$ Both are read "f of g of x"

read "f of g of x"

read f circle g of x

Rewrite as $f(g(x))$ because

Find $g(x)$ stick that into the function

Ex. If $f(x) = x - 5$ and $g(x) = x^2 + 1$, find:

Ex1 $(f \circ g)(-3)$.

1. Rewriting the notation

$$f(g(-3))$$

2. Find $g(-3) = (-3)^2 + 1 = 10$

3. Replace $g(-3)$ with its output into the $f(g(x))$ notation

$$f(10) = 10 - 5 = 5$$

4. Continue

✓ Your turn! $(g \circ f)(-3)$

$$\begin{aligned} &g(f(-3)) \\ &f(-3) = -3 - 5 = -8 \\ &g(-8) = (-8)^2 + 1 = 65 \end{aligned}$$

Practice! Given the definition of $f(x)$ and $g(x)$ below, find the value of:

$$f(x) = 3x - 10. \quad g(x) = x^2 + 2x - 10$$

1. $f(g(2))$

$$\begin{aligned} g(2) &= 2^2 + 2(2) - 10 \\ &= 4 + 4 - 10 = -2 \end{aligned}$$

$$\begin{aligned} f(-2) &= 3(-2) - 10 \\ &= -16 \end{aligned}$$

2. $(f \circ g)(n)$

$$\begin{aligned} &f(g(n)) \\ &f(n^2 + 2n - 10) \\ &3(n^2 + 2n - 10) - 10 \\ &3n^2 + 6n - 40 \end{aligned}$$

Restriction the Domain (possible values of x)

One of those restriction is we can't divide by _____ (makes the function _____)

Many functions have a domain of all real #'s, like $f(x) = 3x^2 - 4$

Rational (fractional) functions are where you need to be careful. 2 situations where not all real #'s.

① fractions! $\frac{\text{num}}{\text{denom}}$ Denom $\neq 0$

② sq roots! $\sqrt{\text{_____}}$ $\text{_____} \geq 0$ b/c can't take sq root of negative #'s.

Find the domain:

1. $v(x) = \frac{x+5}{4x+2}$

Denom $\neq 0$
 $4x+2 \neq 0$
 $4x \neq -2$
 $x \neq -\frac{1}{2}$

2. $k(x) = \frac{x-1}{x^2-12x-28}$

$= \frac{(x-1)}{(x-14)(x+2)}$

Denom $\neq 0$
 $(x-14)(x+2) \neq 0$
 $x-14 \neq 0$ | $x+2 \neq 0$
 $x \neq 14$ | $x \neq -2$

Interval Notation
 left = smallest
 $(-\infty, -2) (-2, 14) (14, \infty)$

You! $q(x) = \frac{x+3}{x^2-14x+40}$

Denom $\neq 0$

$(x-4)(x-10) \neq 0$
 $x-4 \neq 0$ | $x-10 \neq 0$
 $x \neq 4$ | $x \neq 10$

write in Interval Notation
 $(-\infty, 4) (4, 10) (10, \infty)$

exclude 4

You! $f(x) = 2x$

All real #'s
 $(-\infty, \infty)$

$-\infty \leftarrow \rightarrow \infty$

[include # in solution set
 (Don't include

3) $\sqrt{3x+4} = y$
 $3x+4 \geq 0$
 $3x \geq -4$
 $x \geq -\frac{4}{3}$

4) $\sqrt{x-7}$
 $x-7 \geq 0$
 $x \geq 7$
 $[7, \infty)$

Deerz!
 $g(x) = \frac{x+1}{\sqrt{x-2}}$
 combine both restrictions
 $x-2 > 0$
 $x > 2$
 because in denom
 $x-2 \neq 0$